

CPTG: Compact-Quiescent Redshift Stress Test of ZF-COS-20115 at $z = 3.715$

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Abstract

ZF-COS-20115, also reported as ZF-COSMOS-20115, is a compact massive quiescent galaxy observed at redshift $z \simeq 3.715$. The published MOSFIRE measurement gives an integrated stellar velocity dispersion of $283 \pm 52 \text{ km s}^{-1}$, an effective radius of $0.66 \pm 0.08 \text{ kpc}$, a Sérsic index of 4.10 ± 0.99 , and a dynamical mass of $(0.72 \pm 0.27) \times 10^{11} M_{\odot}$. This article uses the object as a compact-quiescent Curvature Polarization Transport Gravity (CPTG) redshift stress test. The effective-radius support scale is computed from the observed stellar dispersion and radius, then divided by the expansion factor at the observed epoch. The midpoint calculation gives a local support scale of $3.9326 \times 10^{-9} \text{ m s}^{-2}$, which normalizes to $6.7739 \times 10^{-10} \text{ m s}^{-2}$ under the Planck-consistent expansion normalization. The result supports the CPTG time-scaled compact-quiescent acceleration branch in the limited effective-radius sense. It is not a full radial dynamical validation, because the available data do not provide a resolved stellar kinematic field.

1 Motivation

Compact quiescent galaxies at high redshift provide a direct way to test whether a local stellar support scale changes coherently with cosmic time. These systems are useful because their stellar bodies are small, massive, and dynamically hot. Their observed velocity dispersions and effective radii therefore define a compact acceleration scale without requiring a resolved rotation curve.

The target is useful for a limited CPTG [6, 7] compact-quiescent stress test. The original discovery paper reported a massive quiescent galaxy at $z = 3.717$ with little or no ongoing star formation [2]. Follow-up work separated the quiescent source from a nearby dusty companion [3]. The MOSFIRE velocity-dispersion study then provided the measurements used below, including the stellar velocity dispersion, effective radius, dynamical mass, and stellar mass scale [1]. The adopted kinematic redshift is

$$z = 3.715. \tag{1}$$

The object has also been used in high-redshift formation and quenching studies, including a model-comparison analysis of early star formation in ZF-COSMOS-20115 [4]. Those formation and SFR quantities are useful context, but they are not used as source terms in the present calculation. The same published kinematic measurements allow a narrow CPTG stress test: compute the effective-radius support scale, normalize it by the expansion factor at the observed epoch, and determine whether the result lies on a finite present-equivalent compact-galaxy branch.

This paper is intentionally limited. It does not attempt a full radial mode reconstruction, a full Jeans solution, or a lensing calculation. It asks a single redshift-stress question: whether the high local acceleration scale of the galaxy at the observed epoch reduces to a stable present-equivalent scale after expansion normalization.

2 Observed Inputs

The calculation uses only the scalar quantities required for a compact effective-radius redshift stress test. The observed inputs are summarized in Table 1 and are taken from the published ZF-COS-20115 MOSFIRE analysis, with discovery and deblending context from the companion observational papers [1, 2, 3].

Observed quantity	Adopted value	Role in stress test
Galaxy	ZF-COS-20115	Target identification
Redshift	$z = 3.715$	Epoch and normalization
Stellar velocity dispersion	$283 \pm 52 \text{ km s}^{-1}$	Pressure-supported anchor
Effective radius	$0.66 \pm 0.08 \text{ kpc}$	Effective-radius scale
Sérsic index	4.10 ± 0.99	Virial-coefficient check
Dynamical mass	$(0.72 \pm 0.27) \times 10^{11} M_{\odot}$	Mass comparator
Stellar mass	$1.15^{+0.16}_{-0.09} \times 10^{11} M_{\odot}$	Photometric check

Table 1: Observed inputs used in the compact-quiescent CPTG redshift stress test of ZF-COS-20115.

The source paper reports the scalar dynamical mass and deblended photometric stellar mass used below [1]. Their central values are not identical, with $M_{\text{dyn}}/M_{\star} \simeq 0.63$, so the mass comparison is treated here as an input-system caveat rather than an independent CPTG mass validation. This is still useful for the present calculation because the published dynamical mass is derived from the same measured scalar kinematic quantities used in the effective-radius stress test. The current star-formation limits and early formation requirements are retained as observational constraints on the object, but no CPTG SFR-output formula is used in this paper.

3 Compact Effective-Radius Support Scale

The local support scale is computed from the measured stellar velocity dispersion and effective radius:

$$a_\sigma = \frac{\sigma_\star^2}{R_e}. \quad (2)$$

Using

$$\sigma_\star = 283 \text{ km s}^{-1}, \quad (3)$$

and

$$R_e = 0.66 \text{ kpc}, \quad (4)$$

gives

$$R_e = 2.0365 \times 10^{19} \text{ m}, \quad (5)$$

and therefore

$$a_\sigma = 3.9326 \times 10^{-9} \text{ m s}^{-2}. \quad (6)$$

This is the observed compact effective-radius support scale at the emission epoch. It is not yet the present-equivalent CPTG stress-test scale because the object is observed at high redshift. It is also not inherited from another object; it is recomputed directly from the observed ZF-COS-20115 dispersion and radius.

4 Expansion Factor and Cosmic Time

The comparison clock is normalized with the Planck 2018 base-parameter reference layer [5]. Thus H_0 is treated explicitly as the Planck-measured Hubble constant used for the dimensional time conversion:

$$H_0^{\text{Planck}} = 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (7)$$

The dimensionless expansion-shape parameters are taken from the same reference layer:

$$\Omega_m = 0.315, \quad (8)$$

and

$$\Omega_\Lambda \simeq 0.685. \quad (9)$$

The expansion factor is

$$E(z) = \sqrt{\Omega_m(1+z)^3 + \Omega_\Lambda}. \quad (10)$$

In this expression, Ω_m and Ω_Λ denote the Planck reference values listed above.

The corresponding dimensional Hubble rate is

$$H(z) = H_0^{\text{Planck}} E(z). \quad (11)$$

At the observed redshift,

$$E(3.715) = 5.8055, \quad (12)$$

so that

$$H(3.715) = 391.29 \text{ km s}^{-1} \text{ Mpc}^{-1}. \quad (13)$$

The corresponding cosmic age is

$$t(z = 3.715) = 1.677 \text{ Gyr}, \quad (14)$$

with a lookback time of

$$t_{\text{lookback}} = 12.119 \text{ Gyr}. \quad (15)$$

Thus the galaxy is observed when the Universe was approximately 12.2 per cent of its present age in this reference cosmology.

5 Redshift-Normalized CPTG Stress Scale

The CPTG redshift stress scale is defined by normalizing the local compact support scale by the expansion factor at the observed epoch, following the compact effective-radius stress-test convention:

$$a_{\sigma,z} = \frac{a_{\sigma}}{E(z)}. \quad (16)$$

Substituting the midpoint values gives

$$a_{\sigma,z} = \frac{3.9326 \times 10^{-9}}{5.8055} = 6.7739 \times 10^{-10} \text{ m s}^{-2}. \quad (17)$$

In the stress-test interpretation, the local emission-time branch is

$$a_{\star}(t_{\text{em}}) = a_{\sigma}, \quad (18)$$

while the present-equivalent branch is

$$a_{\star,0} = a_{\sigma,z}. \quad (19)$$

Therefore

$$a_{\star}(t_{\text{em}}) = 3.9326 \times 10^{-9} \text{ m s}^{-2}, \quad (20)$$

and

$$a_{\star,0} = 6.7739 \times 10^{-10} \text{ m s}^{-2}. \quad (21)$$

The corresponding CPTG curvature radius is tied directly to the same acceleration scale in the CPTG curvature-transport normalization:

$$R_C(t) = \frac{48\pi c^2}{a_{\star}(t)}. \quad (22)$$

The present-equivalent curvature radius is therefore

$$R_{C,0} = 2.0007 \times 10^{28} \text{ m}, \quad (23)$$

while the emission-time curvature radius is

$$R_C(t_{\text{em}}) = 3.4463 \times 10^{27} \text{ m}. \quad (24)$$

The higher early-time acceleration branch therefore corresponds to a smaller curvature radius, while the present-equivalent branch has a lower acceleration scale and a larger curvature radius.

6 Time Branch

Table 2 shows the smooth time branch associated with the midpoint stress-test scale. The table is not a merger history and is not a reconstructed evolutionary path for an individual descendant. It only shows how the same present-equivalent branch maps into different cosmic epochs under the adopted expansion scaling.

Redshift	Cosmic age	$E(z)$	$a_{\star}(t)$	$R_C(t)$
0.00	13.80 Gyr	1.000	6.77×10^{-10}	$2.00 \times 10^{28} \text{ m}$
1.00	5.85 Gyr	1.790	1.21×10^{-9}	$1.12 \times 10^{28} \text{ m}$
3.715	1.68 Gyr	5.805	3.93×10^{-9}	$3.45 \times 10^{27} \text{ m}$

Table 2: Smooth CPTG time branch implied by the midpoint compact-quiescent redshift stress test.

The time branch makes the redshift interpretation explicit. The galaxy is observed on a stronger early-time acceleration branch. Once normalized to the present-equivalent scale, the branch remains finite and compact-quiescent in character.

7 Dynamical-Mass Consistency Check

The source paper reports the following dynamical and stellar mass scales [1]:

$$M_{\text{dyn}} = (0.72 \pm 0.27) \times 10^{11} M_{\odot}, \quad (25)$$

and

$$M_{\star} = 1.15^{+0.16}_{-0.09} \times 10^{11} M_{\odot}. \quad (26)$$

The ratio of the central values is

$$\frac{M_{\text{dyn}}}{M_{\star}} \simeq 0.63. \quad (27)$$

This ratio is below unity, so the mass comparison is treated here only as an input-system caveat and not as an independent CPTG mass validation.

The same midpoint observables also define a raw effective-radius mass scale:

$$M_{\sigma R} = \frac{\sigma_{\star}^2 R_e}{G} \simeq 1.23 \times 10^{10} M_{\odot}. \quad (28)$$

This is lower than the quoted dynamical mass because a pressure-supported compact galaxy is not converted from dispersion and effective radius by the unit coefficient alone. The published dynamical estimate corresponds to an effective structural virial coefficient

$$K = \frac{GM_{\text{dyn}}}{\sigma_{\star}^2 R_e} = \frac{M_{\text{dyn}}}{M_{\sigma R}} \simeq 5.87. \quad (29)$$

This coefficient is not a new CPTG parameter. It represents the ordinary geometric, projection, aperture, anisotropy, and stellar-profile factors that enter a dynamical mass estimate for a pressure-supported system. The value is also consistent with the Sérsic-dependent virial coefficient used by the source paper,

$$\beta(n) = 8.87 - 0.831n + 0.0241n^2, \quad (30)$$

which gives

$$\beta(4.10) = 5.8680. \quad (31)$$

This consistency check should not be read as a CPTG mass validation. It is a sanity check on the input system. The CPTG calculation performed here tests the redshift-normalized effective-radius acceleration branch, not a full mass decomposition.

8 Uncertainty Bracket

The dominant observational uncertainty comes from the stellar velocity dispersion. The published value is [1]

$$\sigma_{\star} = 283 \pm 52 \text{ km s}^{-1}. \quad (32)$$

The effective-radius uncertainty is [1]

$$R_e = 0.66 \pm 0.08 \text{ kpc}. \quad (33)$$

Using the independent extrema gives a broad redshift-normalized range:

$$4.03 \times 10^{-10} \leq a_{\sigma,z} \leq 1.08 \times 10^{-9} \text{ m s}^{-2}. \quad (34)$$

Table 3 shows the central value and bounding cases.

Case	σ_*	R_e	$a_{\sigma,z} \text{ (m s}^{-2}\text{)}$
Low bound	231 km s^{-1}	0.74 kpc	4.03×10^{-10}
Midpoint	283 km s^{-1}	0.66 kpc	6.77×10^{-10}
High bound	335 km s^{-1}	0.58 kpc	1.08×10^{-9}

Table 3: Effective-radius redshift stress scale for the central value and uncertainty extrema.

The midpoint value is the reference calculation. The range should be treated as an uncertainty bracket, not as a fitted interval.

9 Interpretation

The target passes the compact-quiescent CPTG redshift stress test in the limited effective-radius sense. Its observed support scale at redshift

$$z = 3.715 \quad (35)$$

is high, as expected for a compact massive quiescent galaxy. After expansion normalization, however, the scale becomes

$$a_{\sigma,z} = 6.7739 \times 10^{-10} \text{ m s}^{-2}. \quad (36)$$

The key point is not that the local acceleration is small. It is not. The key point is that the high local acceleration at the emission epoch reduces to a plausible present-equivalent compact-galaxy branch when divided by the expansion factor.

The source papers argue that ZF-COS-20115 is a massive, compact, quiescent system at very high redshift, with the integrated stellar velocity dispersion and effective radius required for this scalar stress test [2, 3, 1]. The CPTG stress-test result is consistent with that picture. The object sits on a stronger early-time acceleration branch, while its present-equivalent value remains finite and comparable to the compact high-redshift stress-test scale.

10 Data Needed for a Stronger Test

The present calculation is limited by the available scalar observables reported for ZF-COS-20115 [1]. A stronger CPTG test would require spatially resolved stellar kinematics. The most important missing quantities are

$$\sigma_{\star}(R), \quad (37)$$

and

$$V_{\star}(R), \quad (38)$$

across the stellar body. A full radial test would also require a resolved mass profile, a consistent aperture correction for the measured dispersion, and a clear separation between pressure support and any ordered rotation.

With those data, the object could be tested using a proper radial Jeans reduction or a full CPTG radial field reconstruction. Without them, the appropriate classification is a compact-quiescent redshift stress test, not a full dynamical validation.

11 Technical Summary

The observed midpoint quantities are taken from the published ZF-COS-20115 measurement [1]:

$$\sigma_{\star} = 283 \text{ km s}^{-1}, \quad (39)$$

and

$$R_e = 0.66 \text{ kpc}. \quad (40)$$

They give

$$a_{\sigma} = 3.9326 \times 10^{-9} \text{ m s}^{-2}. \quad (41)$$

At the observed redshift,

$$E(3.715) = 5.8055. \quad (42)$$

Therefore the redshift-normalized stress scale is

$$a_{\sigma,z} = 6.7739 \times 10^{-10} \text{ m s}^{-2}. \quad (43)$$

The corresponding curvature radii are

$$R_C(t_{\text{em}}) = 3.4463 \times 10^{27} \text{ m}, \quad (44)$$

and

$$R_{C,0} = 2.0007 \times 10^{28} \text{ m}. \quad (45)$$

The stress-test outcome is positive in the limited effective-radius sense. It supports the CPTG time-scaled compact-quiescent acceleration branch, while leaving full radial validation for future resolved data.

12 Conclusion

ZF-COS-20115 provides a clean high-redshift target for the compact CPTG stress-test sequence. Its measured dispersion, effective radius, and published dynamical mass are internally consistent under the same Sérsic-dependent virial correction, while the deblended stellar mass remains close enough to serve as a photometric context check rather than a CPTG mass validation. The local effective-radius support scale is high, but after redshift normalization it becomes

$$a_{\sigma,z} = 6.7739 \times 10^{-10} \text{ m s}^{-2}. \quad (46)$$

This value supports the CPTG redshift stress-test branch for compact quiescent galaxies. The result should be stated carefully: CPTG passes this redshift stress test in the effective-radius sense for ZF-COS-20115, but the available data do not constitute a full radial dynamical validation.

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